VARUN VADDI

varunvaddi30600@gmail.com | Grand Rapids, MI | (713) 539-6996 | LinkedIn | Portfolio

SUMMARY

Data Engineer (MS, Data Science) with 5 years building production data pipelines, lakehouse architectures, and ETL systems across Azure and AWS - hands-on across streaming (Kafka), batch (Spark, dbt), orchestration (Airflow), and cloud infrastructure. Most effective when data directly enables a product decision or operational outcome.

SKILLS

Data Engineering: Python, **SQL**, **PySpark**, **Spark**, **Snowflake**, **Databricks**, **dbt**, **Airflow**, **Kafka**, **Delta Lake**, **Parquet**, **REST APIs**, Dimensional Modeling, Lakehouse Architecture, ETL/ELT, Data Quality, Data Governance, CI/CD, **Git**.
Cloud: Azure (ADF, ADLS Gen2, Synapse Analytics, Databricks, Cosmos DB, Azure DevOps), **AWS** (S3, Glue, Athena, Redshift, EMR, SageMaker, EventBridge, SNS), **GCP** (BigQuery, Vertex AI), **Docker**, **Kubernetes**, **GitHub Actions**, **Power BI**, **Tableau**.
Analytics: Statistical Analysis, Feature Engineering, **Dimensional Modeling**, Slowly Changing Dimensions, **Data Quality** Monitoring, Data Profiling, Medallion Architecture, Lakehouse Design, Performance Tuning, Grafana, Matplotlib.

EDUCATION

University of Houston | *M.S. in Engineering Data Science, **GPA**: 3.70* *Dec 2025*

Jawaharlal Nehru Technological University | *B.Tech. in Electronics & Communications Engineering, **GPA**: 3.68* *Jul 2021*

WORK EXPERIENCE

Data Engineer, Meijer, MI *May 2026 - present*

- Building and maintaining **Azure ADF** pipelines with embedded **Databricks** notebooks to ingest Adobe Analytics clickstream data into **ADLS Gen2**, transform via **PySpark**, and deliver curated datasets supporting MarTech analytics and campaign performance reporting in **Power BI**.
- Implementing data quality frameworks using **PySpark/SQL** scripts in Databricks and **Azure DevOps** CI/CD - enforcing null checks, schema validation, and referential integrity across marketing pipelines, ensuring accuracy of media performance metrics for downstream reporting.
- Leading Adobe Analytics-to-in-house site analytics migration - designing logging solutions and **ETL** pipeline architecture that eliminate third-party dependency, debugging ADF-to-Databricks execution failures via Azure Monitor logs and resolving schema drift from upstream Adobe feeds.

Data Science Intern, Wild Genomics, CA *May 2025 - Aug 2025*

- Architected multi-stage **Python** + **Biopython** ETL pipeline ingesting raw **FASTQ** sequencer files - automating demultiplexing, quality filtering, and 97%-identity **clustering** to increase analysis-ready yield from 30% to 80% and cut batch runtime by 40%.
- Built **BLAST**-based classification pipeline automating taxonomic labeling at 95% identity threshold - improved labeling accuracy from 85% to 95% and cut retraining cycle time 3x, enabling weekly **ML** model releases instead of monthly.
- Presented weekly data-driven findings to CEO and CSO, translating complex genomic pipeline outputs into actionable conservation decisions - directly informing 2 strategic outcomes including early detection of bacterial presence across a ranch ecosystem.

Data Analyst, University of Houston, TX *Jul 2024 - Dec 2025*

- Designed **Oracle**-based ETL pipelines integrating three source systems - Common App application data, **Slate CRM** behavioral records, and test score APIs - using rule-based validation and incremental loading, consolidating fragmented pipelines into a unified analytical layer achieving 99%+ data reliability across 40K+ annual applications.
- Built self-service **Tableau** dashboards consuming Oracle analytical tables - surfacing predicted admit rates, yield risk scores, and enrollment funnel KPIs for leadership; automated previously manual workflows reducing admissions officer workload by 70% and accelerating decisions by 5 days.

- Partnered with admissions stakeholders to gather requirements, document data flows across 5+ source systems, and translate business needs into technical pipeline specifications - maintaining data governance standards and ensuring data privacy compliance across all analytical outputs.

Software Engineer, Accenture, India *Sep 2021 - Dec 2023*

- Integrated **Marketo** and **D&B** lead sources into **Salesforce Sales Cloud** via **OAuth 2.0** connected apps - building field mappings and **Apex**/Process Builder transformation logic to land 80M+ CRM records into a unified lead-to-revenue analytical model tracking pipeline health from first touch through closed-won.
- Refactored 30+ batch **Apex** classes and event-driven jobs to **Platform Events** + Queueable async pattern - eliminating governor limit errors under high-volume load, improving ETL throughput 30%, and enabling real-time **Tableau** dashboards that raised conversion 10% (+$2.5M annual impact).
- Built **CRM Analytics (Einstein Analytics)** dashboards and **Tableau** reports using **SOQL**-powered datasets - enforced data access via profiles, permission sets, and field-level security; deployed weekly via CI/CD **changesets** with documented rollback procedures; adopted as single source of truth contributing to $25M revenue growth.

Data Analyst Intern, Capgemini, India *Apr 2021 - Jul 2021*

- Built full-stack internal operations reporting app using **Java Spring Boot** REST APIs, **Angular** frontend, and **MySQL** normalized schemas - designing indexed relational tables with stored procedures, reducing query response times by 30% and improving downstream data reliability by 25%.
- Designed **MySQL** relational database schema for operational data management - defining primary/foreign keys, indexing high-cardinality columns, and writing batch stored procedures supporting transactional data processing for downstream analytics pipelines.

TECHNICAL PROJECTS

Retail Analytics Lakehouse | *Azure Databricks, Delta Lake, Spark, MLflow*

- Built fault-tolerant **Spark** ETL on **Azure Databricks** ingesting 10.5M+ records into **Bronze-Silver-Gold Delta Lake** - SCD Type 2 in Silver, Z-ordering on Gold achieving 5–10x query latency reduction, 100% data quality via automated validation.
- Engineered 45+ behavioral features using **PySpark** window functions and pandas UDFs - trained 5 **MLflow**-tracked models: reorder prediction (87% AUC), churn (77% AUC), LTV (90% R^2); **Hyperopt** tuning +2% AUC; quantified $4.65M annual opportunity via **Streamlit** dashboard.

Hybrid Streaming & Batch Pipeline | *Snowflake, Kafka, dbt, Airflow*

- Engineered **Kafka**-to-**Snowflake** real-time and batch pipeline processing 8.6M NYC taxi records - partitioned topics by borough for parallel ingestion, configured Snowpipe for micro-batch loading, orchestrated **dbt** transformations via Dockerized **Airflow**; 37+ dbt tests at 99.88% accuracy, runtime cut from 6 hours to 30 seconds.
- Built dimensional data marts (fact_trips, dim_location, dim_payment) and **Tableau** dashboards tracking $180M+ quarterly revenue - surfacing 30% peak-hour demand patterns and flagging 3,400+ anomalies via automated quality monitoring.

AWS End-to-End Data Pipeline | *AWS, Athena, Grafana, SageMaker*

- Built batch ETL on **AWS** (S3, Glue, Athena, EventBridge, SNS) processing 12.3M records through Bronze-Silver-Gold architecture - Glue bookmarks for incremental loading, decade-partitioned **Parquet** delivering 10x compression, 99%+ data quality.
- Engineered 43 ML features and iterated **XGBoost** across 4 versions improving R^2 117% - deployed real-time **SageMaker** endpoint under 100ms latency with EventBridge CI/CD, **Grafana** drift monitoring, and SNS alerting.